

DAY 5 WEEK 1

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PROJECT: Battery SoC Estimation & Remaining Charging-Time Prediction Using Machine Learning

Day 5: Pipeline & Experiment Planning

1. End-to-End Pipeline

- 1. Raw Dataset** — Load the LG 18650HG2 CSV data (time-series battery data).
- 2. Inspection** — Check columns, data types, time info, temperature conditions, missing values and unusual records.
- 3. Cleaning** — Handle missing values, duplicates, noise and invalid readings.
- 4. EDA** — Study relationships between features and target variables using graphs and statistics.
- 5. Feature Prep** — Select useful features and create derived features (e.g., power, energy, etc.).
- 6. Session Split** — Separate data by test session to avoid data leakage between train and test sets.
- 7. Baseline** — Train Linear Regression as a simple reference model.
- 8. RF / XGBoost** — Train Random Forest and XGBoost regression models and compare with baseline.
- 9. Evaluation** — Use MAE, RMSE and R^2 for both SoC and remaining charging time.
- 10. Error Analysis** — Check where predictions are wrong (SoC ranges, temperatures, test conditions).
- 11. Final Model** — Choose the best performing and technically suitable model for deployment/use.

2. Experiment Plan

ID	Experiment	Purpose
E1	Mean / Linear baseline	Establish baseline
E2	Random Forest	Nonlinear tree ensemble
E3	XGBoost	Boosted nonlinear model
E4	ANN (optional)	Flexible nonlinear comparison
E5	Best-model tuning	Improve shortlisted model

3. Experiment Log Template

Exp.	Features	Model	Params	MAE	RMSE	R ²	Notes
E1	TBD	Linear Regression	TBD	TBD	TBD	TBD	Establish baseline
E2	TBD	Random Forest	TBD	TBD	TBD	TBD	Nonlinear tree ensemble
E3	TBD	XGBoost	TBD	TBD	TBD	TBD	Boosted nonlinear model
Exp.	Features	Model	Params	MAE	RMSE	R ²	Notes
E4	TBD	ANN (Optional)	TBD	TBD	TBD	TBD	Flexible nonlinear comparison
E5	TBD	Best-model tuning	TBD	TBD	TBD	TBD	Improve shortlisted model

4. Project Folder Structure

```
Battery_SoC_Project/  
|  
|-- data/  
|   |-- raw/  
|       |-- Dataset_Li-ion/  
|           |-- 0degC/  
|           |-- 10degC/  
|           |-- 25degC/  
|           |-- 40degC/  
|           |-- n10degC/  
|           |-- n20degC/  
|       |  
|       |-- processed/  
|       |-- metadata/  
|           |-- data_dictionary.csv  
|  
|-- notebooks/  
|   |-- 01_data_inspection.ipynb  
|   |-- 02_data_quality_eda.ipynb  
|   |-- 03_preprocessing.ipynb  
|   |-- 04_feature_engineering.ipynb  
|   |-- 05_baseline.ipynb
```

```
|  |-- 06_model_comparison.ipynb
|
|-- src/
|  |-- data_loader.py
|  |-- preprocessing.py
|  |-- labels.py
|  |-- features.py
|  |-- split.py
|  |-- models.py
|  |-- evaluation.py
|
|-- results/
|  |-- metrics/
|  |-- models/
|
|-- figures/
|-- reports/
|-- README.md
|-- requirements.txt
|-- .gitignore
```